Inverted MAML Card 1: Round 20 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 30%/40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±90 LSB

1. Round 20 Summary & Cumulative Comparison (R1–R19)

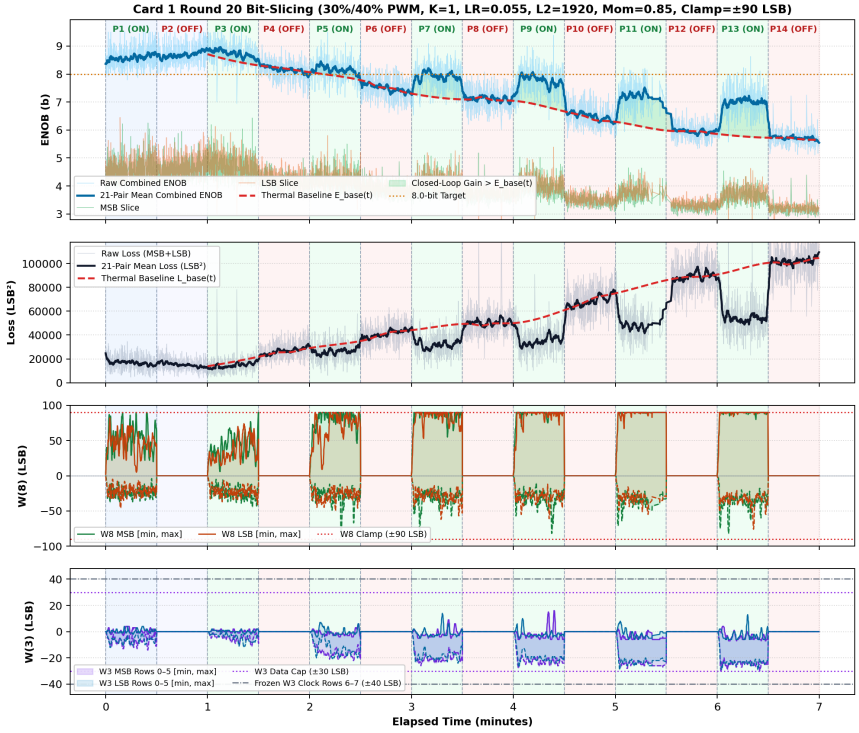
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 20 Detrended Trend Metrics:** Lift vs E_base = **+0.82 b**, Loss Ratio = **1.58x**, Climb = **+0.80 b**, Jitter = **0.41 b**, Recovery = **0.51**, Score = **+1.902**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R12dc_fix	0.055	1920	0.85	±82	+1.04b	1.75x	+0.83b	0.42b	0.00	+1.721
R13dc_fix	0.055	1920	0.85	±85	+1.12b	1.78x	+1.06b	0.43b	0.50	+2.645
R14dc_fix	0.055	1920	0.85	±90	+0.78b	1.55x	+0.97b	0.39b	0.67	+2.117
R15dc_fix	0.055	1920	0.85	±100	+0.94b	1.67x	+0.89b	0.40b	0.75	+2.443
R16dc_fix	0.055	1920	0.85	±90	+0.70b	1.52x	+0.82b	0.34b	0.75	+2.026
R17dc_fix	0.055	1920	0.85	±90	+0.88b	1.65x	+1.25b	0.39b	0.50	+2.385
R18dc_fix	0.055	1920	0.85	±90	+0.77b	1.52x	+0.74b	0.37b	0.67	+1.888
R19dc_fix	0.055	1920	0.85	±90	+0.82b	1.58x	+0.80b	0.41b	0.51	+1.902

2. Phase-by-Phase Telemetry (4,702 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	446	8.40->8.67 (8.58b)	-	20906->15183	[90,82]	14	446
P2	Cold / Regler OFF	414	8.67->8.78 (8.69b)	-	17205->13670	[0,0]	0	0
P3	30% Heat / Regler ON	400	8.86->8.60 (8.73b)	+0.24 b	11876->16285	[90,74]	12	400
P4	30% Heat / Regler OFF	382	8.34->8.07 (8.16b)	-0.05 b	22816->28802	[0,0]	0	0
P5	30% Heat / Regler ON	353	8.12->8.09 (8.16b)	+0.25 b	27529->27481	[90,90]	22	353
P6	30% Heat / Regler OFF	350	7.52->7.39 (7.50b)	+0.01 b	39951->43715	[0,0]	0	0
P7	30% Heat / Regler ON	334	7.46->7.89 (7.90b)	+0.78 b	42646->33132	[90,90]	26	334
P8	30% Heat / Regler OFF	320	7.19->7.08 (7.11b)	+0.02 b	48151->51368	[0,0]	0	0
P9	40% Heat / Regler ON	322	7.27->7.64 (7.74b)	+0.89 b	47600->37739	[90,90]	28	322
P10	40% Heat / Regler OFF	315	6.67->6.34 (6.49b)	+0.02 b	63025->73184	[0,0]	0	0
P11	40% Heat / Regler ON	207	6.63->7.27 (7.17b)	+1.18 b	66378->46507	[90,90]	29	207
P12	40% Heat / Regler OFF	284	6.03->5.94 (5.95b)	+0.04 b	85506->90078	[0,0]	0	0
P13	40% Heat / Regler ON	294	6.06->6.98 (6.93b)	+1.25 b	85197->54565	[90,90]	30	294
P14	40% Heat / Regler OFF	281	5.73->5.69 (5.72b)	+0.03 b	100961->103334	[0,0]	0	0


Card 1 Round 20 Bit-Slicing (30%/40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±90 LSB)
P1 (ON)
P2 (OFF)
P3 (ON)
P4 (OFF)
P5 (ON)
P6 (OFF)
P7 (ON)
P8 (OFF)
P9 (ON)
P10 (OFF)
P11 (ON)
P12 (OFF)
P13 (ON)
P14 (OFF)
ENOB (b)
Raw Combined ENOB
21-Pair Mean Combined ENOB
MSB Slice
LSB Slice
Thermal Baseline E_base(t)
Closed-Loop Gain > E_base(t)
8.0-bit Target
Loss (LSB²)
Raw Loss (MSB+LSB)
21-Pair Mean Loss (LSB²)
Thermal Baseline L_base(t)
W(8) (LSB)
W8 MSB [min, max]
W8 LSB [min, max]
W8 Clamp (±90 LSB)
W(3) (LSB)
W3 MSB Rows 0–5 [min, max]
W3 LSB Rows 0–5 [min, max]
W3 Data Cap (±30 LSB)
Frozen W3 Clock Rows 6–7 (±40 LSB)
Elapsed Time (minutes)

3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 20

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±90 LSB, while W(3) rows 0–5 stay within ±30 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 20 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

